BUSINESS REQUIREMENTS DOCUMENT (BRD)

Project Name: EduBridge Nigeria
Document Version: 1.0
Prepared By: Business Analyst
Prepared For: Executive Director, Pan-African Education Foundation, Pilot School Principals, Technology Team
Date: September 2026

1. EXECUTIVE SUMMARY

1.1 Background

Nigeria has more than 10.5 million out-of-school children, the highest number globally, while 68% of government scholarship opportunities are awarded to urban applicants because many rural students lack access to timely information. EduBridge Nigeria is proposed to address these challenges by providing an offline-first digital platform that expands access to learning resources, scholarship opportunities, mentorship, and academic progress tracking for students in underserved communities.

The proposed solution addresses three key educational challenges. First, it improves access to WAEC-aligned learning materials through SMS-enabled downloads and offline content, enabling students to study without continuous internet access despite limited textbook availability. Second, it improves access to scholarship opportunities by matching eligible students with verified funding programmes and sending automated SMS reminders before application deadlines. Third, it strengthens academic support by enabling teachers to record weekly assessment results and allowing students and parents to monitor academic progress through monthly reports, making it possible to identify learning gaps earlier and provide timely intervention.

EduBridge Nigeria is intended to support approximately 25,000 students across five pilot states during a ten-month pilot implementation.

2. PROBLEM STATEMENT

Element	**Operational Context**	**Key metric**
Situation *(Current Reality)*	Nigeria has the largest population of out-of-school children in the world. Students enrolled in rural public secondary schools also face limited access to learning materials and internet-dependent educational resources, reducing opportunities for independent learning.	● 10.5 million children out of school ● • Average of 0.4 textbooks per student per subject

COMPLICATION *(The Systemic Frictions)*	Rural students have limited awareness of scholarship opportunities, limited access to structured career guidance, and parents have minimal visibility into academic progress. Teachers also face increasing workloads with limited digital support.	• 68% of government scholarship awards are received by urban students • No structured framework for career mentoring or parent progress monitoring
IMPACT *(The Compounded Threat)*	Learning gaps are often identified only after end-of-term assessments, limiting opportunities for timely intervention. This contributes to poorer academic outcomes and increases the risk of students leaving school before completing secondary education.	Early intervention can reduce dropout risk by up to 67% compared with end-of-term intervention
QUESTION *(The Core Product Goal)*	How can EduBridge Nigeria provide an offline-capable digital platform that enables rural secondary school students to access curriculum content, discover scholarship opportunities, receive structured mentoring, and monitor academic progress regardless of internet availability?	**Target Outcome:** Improve educational access and reduce dropout risk through timely academic intervention during the pilot programme

3. STAKEHOLDER ANALYSIS

Name/Role	Power/Interest	Primary Need	Key Concern	Engagement Strategy	Communication Frequency.
Dr. Joshua Gideon *(Executive Director & Project*	High Power / High Interest	Documented evidence that the system is actively improving	Low adoption by teachers and students may reduce the programme's	Provide weekly programme performance dashboards, student outcome reports, and adoption metrics.	Weekly via formal dashboard summary.

Sponsor)		student learning outcomes across target schools.	expected educational outcomes.		
Pan-African Education Foundation *(Grant Funder)*	High Power / Low Interest	Evidence demonstrating measurable educational outcomes and programme impact, including verified scholarship applications and measurable learning growth.	Missing impact data across the 6-month and 12-month grant reporting cycles.	Set up formal impact reporting frameworks that track student progress metrics and successfully submit scholarship applications.	Quarterly via formal evaluation reviews.
State Ministries of Education *(5 Pilot States)*	High Power / High Interest	Aggregated student academic performance records for regional and state-level policy reporting.	Confusion or lack of clarity regarding who holds legal ownership of student data.	Deploy local data storage protocols and provide explicit data ownership agreements that keep student data under state oversight.	Monthly formal executive briefings.
Secondary School Principals *(50 Pilot Schools)*	High Power / High Interest	A stable educational program that delivers real results and remains active over the long term.	A concern that this is just another short-term program that adds work and disappears after 6 months.	Appoint principals as school-level program heads; provide them with executive usage dashboards to showcase local impact.	Bi-weekly via a web summary dashboard.
Subject Teachers *(750 across 50 Pilot Schools)*	Low Power / High Interest	Time-saving tools that streamline their lesson planning routines rather than increasing their daily workload.	A fear that new technology will simply translate into more hours of unpaid administrative tasks.	Provide a lesson plan library that reduces lesson preparation time by approximately three hours per week, supported by professional development badges and peer recognition.	Weekly via app ledger; Monthly via SMS recognition.
Rural Public Students *(Primary Users, Ages 11–18)*	Low Power / High Interest	Reliable offline access to high-quality WAEC-aligned learning materials without requiring	App failures due to a lack of individual internet connectivity or lack of smartphone access.	Deliver a zero-data, offline-cached Android layout combined with a native USSD gateway (*EDU*DL#).	Daily interactive usage; Weekly SMS motivation pings**.**

		continuous internet connectivity .			
Scholarship Seeking Students *(SS2–SS3 Cohort)*	Low Power / High Interest	Clear information about verified scholarship opportunities that match their eligibility, including application requirements and guidance on how to apply.	Missing scholarship application deadlines because of limited access to timely information and application guidance .	Run automated offline eligibility sorting and push targeted, actionable SMS countdown alerts.	Real-time in-app data; Weekly targeted SMS tip**s.**
Volunteer Professional Mentors *(Doctors, Engineers, Lawyers)*	Low Power / High Interest	A highly organized, time-efficient engagement structure that does not pull them away from their day jobs.	Unstructured mentoring sessions and poor student attendance may discourage continued participation.	Provide structured monthly mentoring sessions, automated reminders, and prepared discussion guides to ensure each session is productive.	Bi-weekly via SMS/WhatsApp; Quarterly feedback loop**s.**
Scholarship Bodies *(MTN, Agbami, NDDC, etc.)*	High Power / High Interest	A pipeline of high-quality, fully completed application packets from qualified rural applicants.	Applicant pools that are heavily dominated by urban students who already hold an informational edge.	Provide a verified pipeline of candidates who have completed mandatory prep modules and submit complete applications at least seven days before the application deadline.	Monthly via automated data insights report
Parents / Guardians	Low Power / High Interest	Clear information about their child's academic performance and progress toward examination readiness.	A concern that mobile device access will distract students from their core textbook studies.	Send an automated, clear monthly summary text on the last Friday of every month detailing grades and milestones.	Monthly via direct automated SMS text

IT Development Team	Low Power / High Interest	Stable requirements and clear documentation to build the core system architecture.	Changing project parameters mid-build, or unrealistic delivery timelines.	Maintain agreed project requirements, prioritised sprint backlogs, and regular sprint review meetings to minimise scope changes during development.	Weekly via interactive sprint reviews.
Nigeria Data Protection Commission (NDPC)	High Power / Medium Interest	Compliance with the Nigeria Data Protection Act (NDPA) and protection of student personal information.	Unauthorised collection or disclosure of minors' personal data.	Conduct periodic compliance reviews, implement privacy controls, and maintain parental consent records.	Quarterly compliance reports.

4. PROJECT SCOPE

4.1 In Scope

Student Features

- Student registration and login
- Student profile management
- Offline WAEC learning content
- SMS/USSD learning content download
- Learning content search
- Scholarship search and eligibility matching
- Scholarship application guidance
- Scholarship deadline SMS notifications
- Mentor matching
- Structured mentorship sessions
- Academic progress tracking
- Learning history

Teacher Features

- Teacher registration and verification
- Teacher profile management
- Weekly assessment entry
- Student performance tracking
- WAEC lesson plan library
- Lesson plan sharing
- Professional achievement badges

Parent Features

- Parent registration
- Student progress dashboard
- Monthly progress SMS notifications

Mentor Features

- Mentor registration and verification
- Mentor profile management
- Mentoring session management
- Session agenda SMS notifications
- Session completion tracking

Scholarship Provider Features

- Scholarship listing management
- Scholarship application tracking
- Scholarship performance reporting

School Administrator Features

- Student management
- Teacher management
- School performance monitoring
- Academic performance reporting

System Administrator Features

- User management
- Student management
- Teacher management
- Parent management
- Mentor management
- Scholarship management
- User verification
- Programme reporting
- State performance dashboard
- SMS gateway monitoring
- System monitoring
- Audit logs

4.2 Out of Scope

OUT OF SCOPE	Justification
Live Video Tutoring	Requires reliable broadband internet, which is not consistently available in the target rural communities. Excluded to maintain the project's offline-first design.
WAEC/JAMB Online Examination Integration	Requires formal partnerships and system integration with examination bodies. Planned for a future release after the pilot implementation.
Student-to-Student Messaging	Excluded to minimise child safeguarding, content moderation, and data protection risks during Version 1.
School Fee Payment Processing	Falls outside the project's core educational objectives and would require integration with licensed payment service providers.

5. Functional Requirement

Module Name	Req ID	Functional Requirement
Offline Content Library	FR-OCL-001	The system shall allow a student to request a lesson download link by dialling the designated USSD shortcode (*EDU*DL#) from any active GSM mobile network.
	FR-OCL-002	The system shall allow a student to download WAEC-aligned revision content for up to 4 core subjects via an SMS download link (*EDU*DL#) to any Android device.
	FR-OCL-003	The application shall ensure that all downloaded study files and practice questions remain accessible in full without requiring any active internet connection or data.
	FR-OCL-004	The system shall store downloaded study content locally on the device directory for a minimum baseline of 90 days before requiring a network

		validation refresh.
	FR-OCL-005	The system shall notify the student when locally stored learning content is due for validation or update after 90 days of offline use and prompt the user to connect to the internet to refresh the content.
Content Search & Navigation	FR-CSN-001	The system shall allow a student to search downloaded content by subject, topic, and WAEC year (2019–2024) without requiring any internet connectivity.
	FR-CSN-002	The system user interface shall allow a student to navigate to any specific study topic in under 3 menu selections from the main dashboard, without requiring internet connectivity
Scholarship Directory	FR-SDB-001	The system shall maintain a centralized database containing a minimum of 50 active, verified corporate, state, and national scholarship portfolios.
	FR-SDB-002	The system database shall log and display the unique scholarship name and the specific funding body for every listed record in the offline registry.
	FR-SDB-003	The system database shall save the exact eligible states, eligible class levels, and annual value in Naira (₦) for every listed scholarship entry.
	FR-SDB-004	The system database shall record the official application deadline, a key document checklist, and a direct link to the application portal or physical submission address
Scholarship Matching	FR-SMT-001	The system shall display to each student a personalized list of scholarships they are eligible for based on their registered state, class level, and subject combination.
	FR-SMT-002	The matching engine shall rank the personalized matching list chronologically, placing the closest application closing deadlines (soonest first) at the very top of the student's feed.
	FR-SMT-003	The application shall update and refresh all local student matching feeds within 24 hours of any global change made to a scholarship's criteria or deadline.
Scholarship Deadline Alerts	FR-SDA-001	The system shall send an automated SMS alert over basic cellular channels to every student who has flagged an upcoming scholarship opportunity as 'interested'.
	FR-SDA-002	The system shall dispatch deadline alerts exactly at 30 days, 14 days, and 3 days before the application

		deadline, including: scholarship name, deadline date, key document checklist, and a 3-step application summary.
Mentor Matching	FR-MMT-001	The system shall match each registered student with up to 3 available volunteer professional mentors using background database filters**.**
	FR-MMT-002	The matching engine shall pair users based on three specific fields: career aspiration (selected from a menu of 15 paths), geographic state of origin, and core subject strength
	FR-MMT-003	The application shall display the mentor's name, active profession, state of origin, and a 3-sentence profile bio for the student to choose from.
Structured Mentoring Sessions	FR-SMS-001	The system shall automatically generate a structured 45-minute meeting agenda for each confirmed mentor-student match.
	FR-SMS-002	The platform shall push the agenda—containing 5 clear milestones (goals review, obstacle discussion, resource sharing, action commitment, and next session goal)—via basic cell SMS to both parties 24 hours prior to the session
Student Progress Tracking	FR-SPT-001	The system shall allow a subject teacher to enter a student's weekly quiz score (0–100) per subject via the web portal in under 2 minutes per student record.
	FR-SPT-002	The portal shall make scores visible to the student immediately within local offline storage and automatically save them to be compiled into the parent's monthly report.
Parent Progress Report (SMS)	FR-PPR-001	The system shall automatically compile and send each registered parent a monthly progress summary via cellular SMS text on the last Friday of every calendar month.
	FR-PPR-002	The monthly parent text report shall summarize: their child's average quiz score by subject this month, number of scholarship alerts received, number of content modules accessed, and whether a mentor session was completed.

6. NON-FUNCTIONAL REQUIREMENTS

Requirement Category	Req ID	Non-Functional Requirement
Offline Capability *(The Primary Constraint)*	NFR-001	The system shall deliver 100% of the core student learning functions (content access, progress review) without any internet connectivity, using locally cached content updated via SMS download link or Wi-Fi synchronization when available, with zero feature degradation in offline mode compared to online mode for core learning features.
	NFR-002	To bypass cellular data costs entirely, the platform shall route 100% of scholarship reminders and mentorship agendas through standard network SMS text channels instead of relying on internet-dependent app notifications
	NFR-003	Downloaded revision content shall remain fully functional on the device storage for a minimum baseline of 90 days before requiring a simple, data-free validation refresh.
Performance	NFR-004	The application shall open and display any downloaded text topic on a low-end Android mobile phone (devices with 1GB RAM or less) in under 4 seconds, even when the phone has zero internet connection.
	NFR-005	The system shall synchronise locally stored student progress data within 5 minutes after the teacher's device reconnects to a Wi-Fi or cellular network.
Security	NFR-006	The system shall encrypt personal data at rest and in transit using AES-256 encryption to comply with the Nigeria Data Protection Act (NDPA) and applicable data protection requirements.
	NFR-007	The system shall prevent any student from accessing another student's progress data or personal profile, verified through unique login credentials with session-level access control.
Usability	NFR-008	A first-time student user in Zamfara, Kebbi, Yobe, Ebonyi, or Cross River with no prior technology training SHALL be able to complete the core tasks (download content, view a scholarship match, check their progress score) unassisted within 10 minutes of first use
	NFR-009	The system shall maintain consistent navigation and terminology across all student screens to minimise user errors during task completion.
Accessibility	NFR-010	The system shall present all learning content and user interface text in English using clear, simple language suitable for students whose first language may be Hausa, Yoruba, or Igbo.
Data Protection	NFR-011	The system shall obtain and record parental or guardian consent

		before collecting or processing personal information for students under 18 years of age in accordance with the Nigeria Data Protection Act (NDPA).
Mentor Safeguarding	NFR-012	The system shall prevent mentors and students under 16 years of age from exchanging personal contact information unless approved by the school principal.
Scalability	NFR-013	The system shall support an initial deployment of 25,000 students across the five pilot states and allow expansion to at least 1,000,000 students without requiring changes to the core system architecture.

7.User Stories with Acceptance Criteria

US-001: As a JSS3 student in Zamfara with no data plan, I want to download my Mathematics revision materials via an SMS link so that I can study at home using my family's phone without needing an internet connection.

- *Acceptance Criteria:* Given a student has zero mobile data, When they dial the shortcode *EDU*DL# for JSS3 Math, Then the system must send an offline lesson link via SMS in under 30 seconds.

US-002: As an SS2 student preparing for WAEC, I want to see a personalized list of scholarships I qualify for ranked by deadline so that I never miss an application window because I found out too late.

- *Acceptance Criteria:* Given an active student profile, When they view the offline directory, Then the app must display eligible scholarships sorted chronologically with the closest closing deadline first.

US-003: As an SS3 student who wants to study Medicine, I want to be matched with a Nigerian doctor who came from a similar background so that I have someone to guide my university application and show me the path is possible.

- *Acceptance Criteria:* Given a student profile selecting "Medicine", When they click match mentor, Then the system must display up to 3 verified mentors from their home state with 3-sentence bios within 60 seconds.

US-004: As a Mathematics teacher at a pilot school, I want to access a library of lesson plans created by other EduBridge teachers so that I can reduce my weekly lesson preparation time and use better-quality materials.

- *Acceptance Criteria:* Given a verified teacher account login, When they search by class and topic, Then the portal must fetch downloadable lesson templates in under 5 seconds.

US-005: As a parent in Ebonyi State, I want to receive a monthly SMS telling me how my child is performing in each subject so that I can support their studies even if I was not educated myself.

- *Acceptance Criteria:* Given the last Friday of the month arrives, When the auto-reporting engine runs, Then the system must push a clear text progress report to the parent's registered basic mobile line.

US-006: As a volunteer mentor (an engineer in Lagos), I want to receive a structured 45-minute session agenda before each mentoring session so that the time is productive and I know exactly what I am committing to.

- *Acceptance Criteria:* Given a confirmed match calendar date, When it is exactly 24 hours before the session, Then the system must text a 5-point session agenda script to both lines via SMS.

US-007: As the Executive Director, I want a programme dashboard showing platform engagement by state, scholarship applications submitted, and mentor sessions completed so that I can report accurate impact data to the foundation.

- *Acceptance Criteria:* Given an administrator account login, When the home dashboard console opens, Then the system must display real-time usage graphs across all 5 pilot states.

US-008: As an MTN Foundation scholarship coordinator, I want to see the number of EduBridge students who flagged my scholarship as 'interested' and how many completed the preparation modules so that I can assess whether EduBridge is improving my applicant pool quality.

- *Acceptance Criteria:* Given a corporate partner login, When they load the pipeline tracker, Then the workspace must display live student tracking counts and module preparation progress percentages.

US-009: As a public school chemistry teacher, I want to enter weekly quiz scores for my students while my phone is offline so that I can log their grades immediately without paying for personal data bundles.

- *Acceptance Criteria:* Given zero network data, When a teacher saves student scores (0-100), Then the app must cache data locally and auto-sync to central servers within 5 minutes of network reconnection.

US-010: As a corporate funding donor board representative, I want to submit a new scholarship directory listing through an upload form so that our organization can reach qualified rural candidates without manual administrative delays.

- *Acceptance Criteria:* Given a complete form submission, When a partner clicks submit, Then the system must hold the file as an unverified draft state until an EduBridge administrator verifies it.

US-011: As an SS3 student from an ineligible state, I want the system to block me from tracking a state-specific funding pool so that I do not spend my time applying for opportunities I do not qualify for.

- *Acceptance Criteria:* Given a regional profile mismatch, When a student tries to flag a state-restricted scholarship, Then the system must deny the flag action and display an explicit plain-language error text.

US-012: As a student who has lost their phone, I want to recover my data and progress tokens using a verification text key so that I can resume studying on a new device without starting over.

- *Acceptance Criteria:* Given a phone number login request, When the user types their original phone number and inputs the received 6-digit text code, Then the system must restore the student's study records associated with the verified account.

US-013: As a program safety tracking officer, I want the platform to flag a finished mentoring session that was not reported by the student within 48 hours so that I can check on user safety and system compliance.

- *Acceptance Criteria:* Given a session marked finished by a mentor, When 48 hours pass without student verification, Then the system must change row status to "Under Review" and text the school principal.

US-014: As a busy professional mentor, I want to cancel an active meeting slot using a short command text so that my student is notified immediately and my time availability is reset for other matches.

- *Acceptance Criteria:* Given a locked calendar date, When a mentor sends a "CANCEL" command text, Then the system must clear the block and push an alert text to the student phone line.

US-015 As a grant manager at the Pan-African Education Foundation, I want to request a 6-month project outcome report so I can verify student progress against our ₦200M funding milestone criteria.

- *Acceptance Criteria:* Given the project tracking timeline hits the 6-month baseline mark, When the manager generates the summary report from the admin page, Then the platform must output verifiable, aggregated datasets detailing active study module downloads and completed scholarship applications.

8. PROCESS OVERVIEW

Current State

Scholarship Process

Student → Hear about scholarship from friend/radio/poster → Request support from parent/guardian→ Travel to scholarship office to verify information/collect application requirement → Visit school for transcript → Visit bank/local government → Gather documents → Go to Cybercafe/paper to complete application→ Possible missed deadline

Issues

- 3–6 week discovery delay.
- Physical travel.
- Long queues.
- Paper-based applications.
- Cybercafe costs.
- Multiple application visits.
- 100–300 km submission distances.
- 43% deadline failure rate among rural applicants who start the process.

Learning Support Process

Student → Identifies that they do not understand a topic → Asks subject teacher for help → Teacher unavailable → Attempts to borrow textbook → Textbook already borrowed → Visits cybercafe for YouTube tutorial → Internet access/cost becomes a barrier → Considers buying revision guide → Cost becomes a barrier → Searches for another available resource → Learning gap may remain unresolved

Issues

- Limited access to textbooks
- Teachers may be unavailable when students need additional support.
- Cybercafe learning resources cost approximately ~~₦~~300–~~₦~~500 per hour.
- Revision guides may cost ~~₦~~2,000–~~₦~~4,500.
- Students may wait **5–14 days** before accessing a reliable learning resource.

Future State (With EduBridge)

Scholarship Process

Scholarship Body → Publish/Update Scholarship on EduBridge → Student Receives SMS Alert → Student Views Scholarship Details → System Checks Eligibility → Student Reviews Requirements → Student Uses Application Guide/Preparation Support → Completes Application → Submits Application through Official Portal → EduBridge Records Application Guidance/Confirmation → Student receives submission confirmation

Learning Support Process

Student → Identifies Knowledge Gap → Opens EduBridge Offline Content Library → Searches Subject/Topic → Selects Learning Resource → Studies Revision Notes/Examples → Completes Practice Exercise/Quiz → Reviews Progress → Learning Gap Addressed

Benefit

- Offline-capable curriculum content library:
- Mentor matching